

- 5 (a) (i) waves at (each) slit/aperture spread B1
hence wave(s) able to superpose/meet/interfere in overlap region B1
- (ii) Unable to achieve constant phase difference/coherence for two separate light source(s) B1
or
to ensure waves/light from the double slit are coherent/have constant phase difference
- (iii) Greater intensity of bright fringes B1
Narrower bright fringes B1
- (b) $x = \lambda D / a$ C1
 $\lambda = (36 \times 10^{-3} \times 0.48 \times 10^{-3}) / (16 \times 2.4)$ C1
 $= 4.5 \times 10^{-7} \text{ m}$ A1
- (c) (i) water is flat/still/no disturbance B1
destructive interference as the path difference is 2.5λ when sources start in phase B1
- (c) (ii) 1. surface/water/P vibrates/oscillates with max amplitude waves from the two dippers arrive in phase B1
2. surface/water/P vibrates/oscillates as a non-zero minima due to incomplete destructive interference B1

